# Probability and Statistic (for CS Students)

## Lecture 13: Continous Random Variables II

University of Mazandaran

Fall 1405

# Recap

- Probability Density Function (PDF) conveys complete information about a continuous random variable.
- For a continuous random variable $X$, the PDF of $X$ is the function $f_X : \mathbb{R} \to \mathbb{R}$, such that:
  - $f_X(z) \geq 0$ for all $z \in \mathbb{R}$: it is always non-negative
  - $\int_{-\infty}^{\infty} f_X(z)\, dz = 1$: the area under the entire curve is 1
  - $P(a \leq X \leq b) = \int_a^b f_X(z)\, dz$: the probability that $X$ lies in $[a, b]$ is the area under the curve from $a$ to $b$
  - $P(X = w) = P(w \leq X \leq w) = \int_w^w f_X(z)\, dz = 0$: the probability of being a particular value is 0
  - $P(X \approx q) = P\left(q - \frac{\epsilon}{2} \leq X \leq q + \frac{\epsilon}{2}\right) \approx \epsilon f_X(q)$

# Recap

What do you get if you integrate over a probability density function?

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**A Probability**

# Exponential Random Variable

Consider an experiment that lasts a duration of time until success occurs.

Def: an exponential random variable $X$ models the amount of time until success.

$$X \sim \text{Exp}(\lambda)$$

$$\text{Support: } [0, \infty)$$

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Examples:

- Time until next earthquake
- Time for request to reach web server
- Time until the next phone call

# Exponential Random Variable

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$$X \sim \text{Exp}(\lambda)$$

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$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

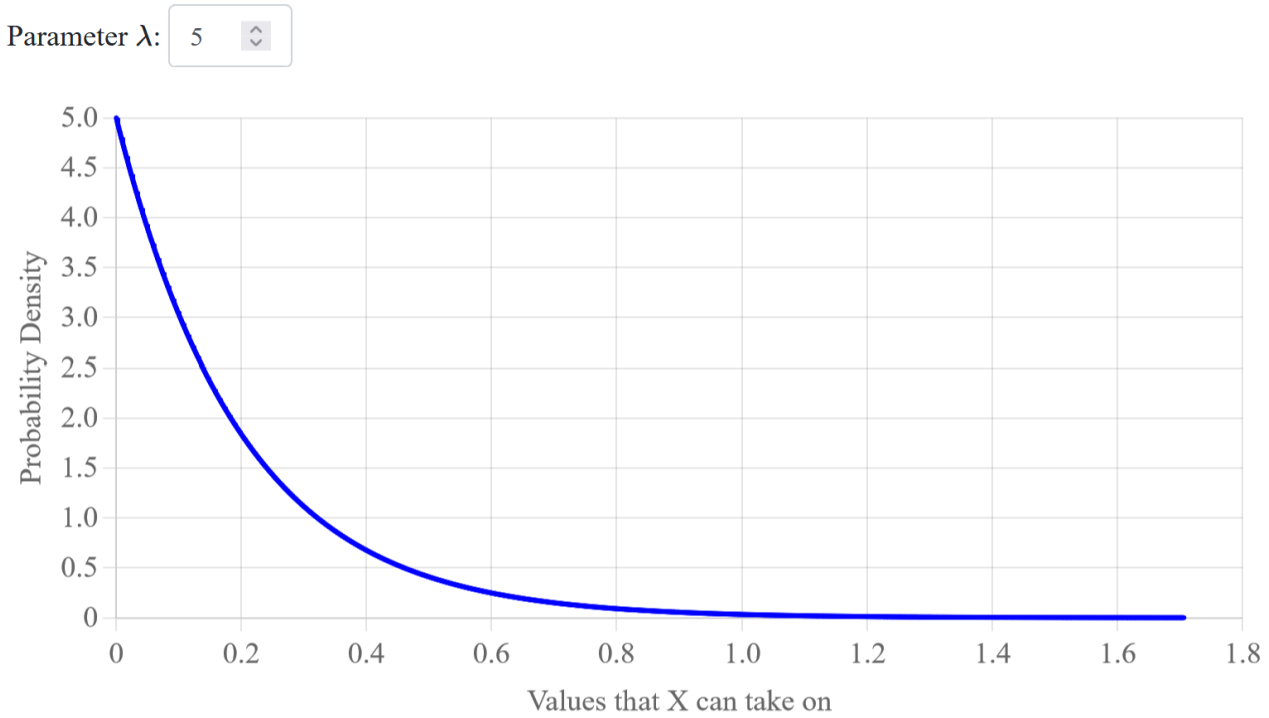
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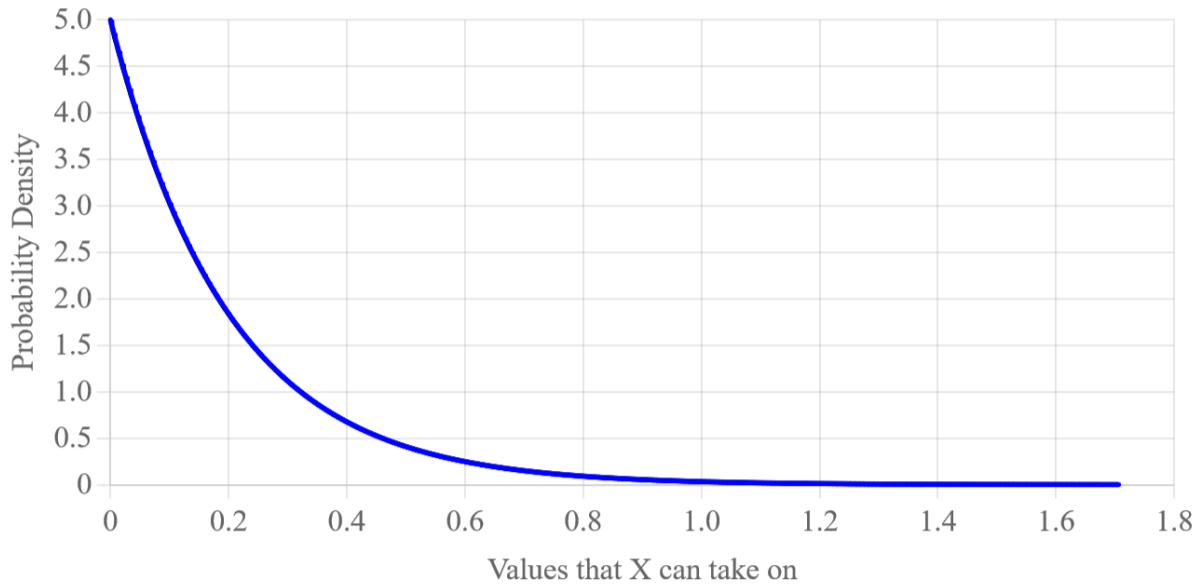


# Exponential Random Variable

| | |
|---|---|
| **Notation:** | $X \sim \text{Exp}(\lambda)$ |
| **Description:** | Time until next events if (a) the events occur with a constant mean rate and (b) they occur independently of time since last event. |
| **Parameters:** | $\lambda \in \mathbb{R}^+$, the constant average rate. |
| **Support:** | $x \in \mathbb{R}^+$ |
| **PDF equation:** | $f(x) = \lambda e^{-\lambda x}$ |
| **CDF equation:** | $F(x) = 1 - e^{-\lambda x}$ |
| **Expectation:** | $\text{E}[X] = 1/\lambda$ |
| **Variance:** | $\text{Var}(X) = 1/\lambda^2$ |

**PDF graph:**

Parameter $\lambda$: 5



The process for an Exponential and a Poisson are the same. So is the paramter $\lambda$. The question is different.

# How Many Earthquakes

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$X =$ Number of major earthquakes next year

$X \sim \text{Poi}(\lambda = 0.002)$

$$P(X = 0) = \frac{\lambda^0 e^{-\lambda}}{0!} = \frac{0.002^0 e^{-0.002}}{0!} \approx 0.998$$

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# Integral Review

$$\int e^{cx} dx = \frac{1}{c} e^{cx}$$

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$= \left(-e^{-0.06} + e^0\right) \approx 0.06$

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$Y \sim \text{Exp}(\lambda = 0.002)$

$E[Y] = \frac{1}{\lambda} = \frac{1}{0.002} = 500$

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$Y \sim \text{Exp}(\lambda = 0.002)$

$\text{Var}(Y) = \frac{1}{\lambda^2} = \frac{1}{0.002^2} = 250,000 \text{ years}^2$

$\text{Std}(Y) = \sqrt{\text{Var}(X)} = 500 \text{ years}$

Is there a way to avoid integrals?

# Cumulative Density Function

A cumulative density function (CDF) is a "closed form" equation for the probability that a random variable is less than a given value

$$F(x) = P(X < x)$$

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$$F(x) = P(X < x)$$

If you learn how to use a cumulative density function, you can avoid integrals!

# CDF of an Exponential

$$F_X(x) = 1 - e^{-\lambda x}$$

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$$F_X(x) = 1 - e^{-\lambda x}$$

$$P(X < x) = \int_{y=-\infty}^{x} f(y)\, dy$$

$$= \int_{y=0}^{x} \lambda e^{-\lambda y} dy$$

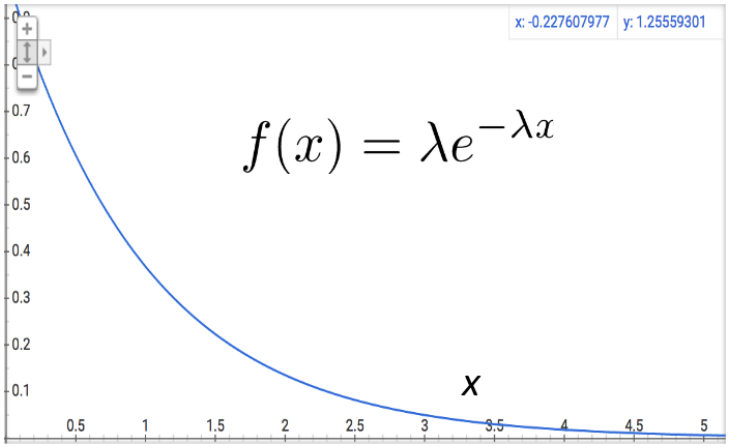
$$= -e^{-\lambda y}\Big|_0^x$$

$$= [-e^{-\lambda x}] - [-e^{\lambda 0}]$$

$$= 1 - e^{-\lambda x}$$

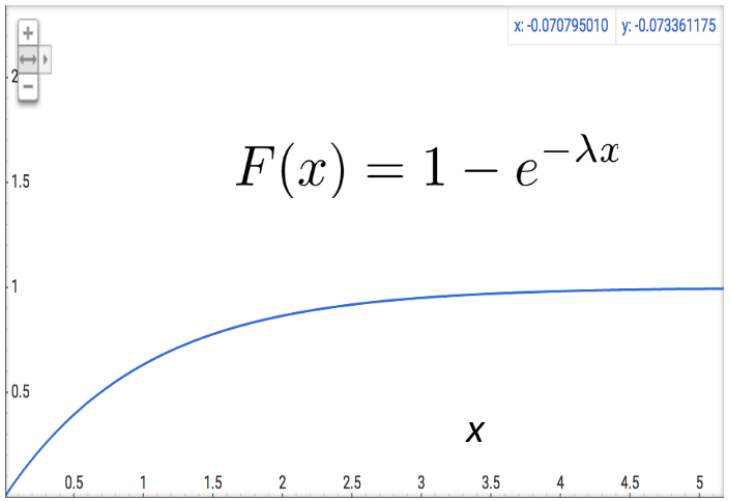
# Using CDF Example. $X$ is Exp($\lambda = 1$)

Probability density function



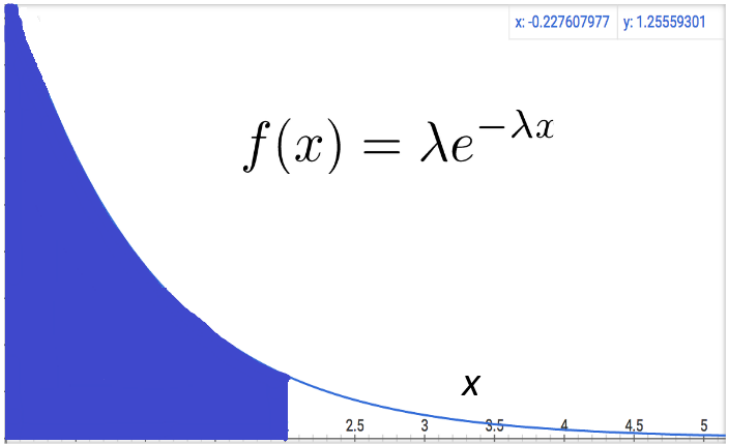

$P(X < 2)$

Cumulative density function

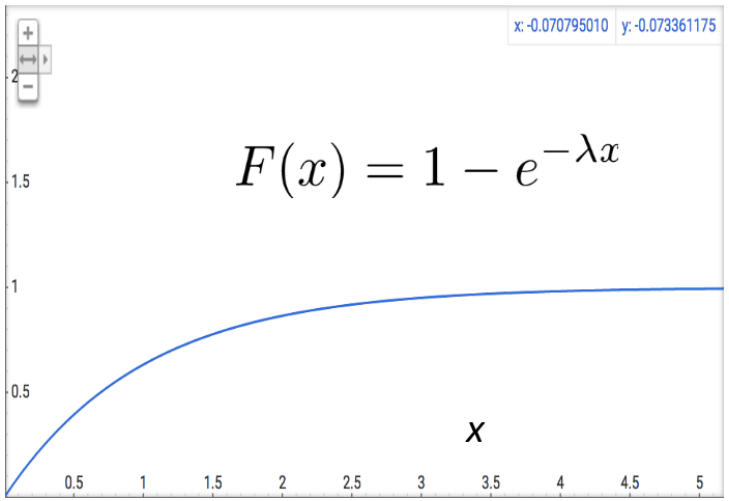


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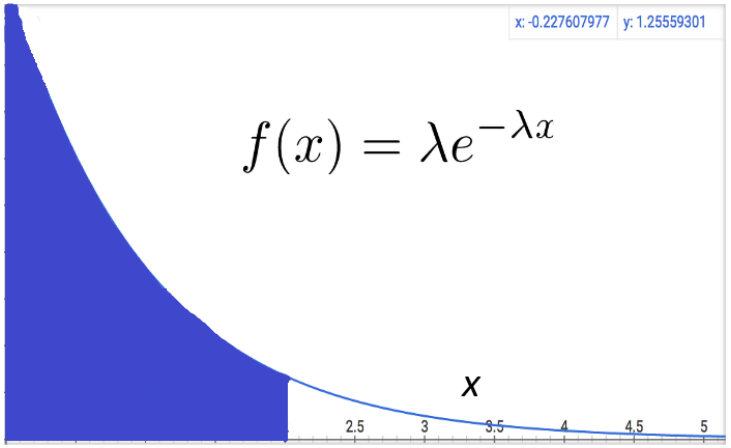



$$P(X < 2)$$

$$= \int_{z=-\infty}^{2} f(x)\, dx$$

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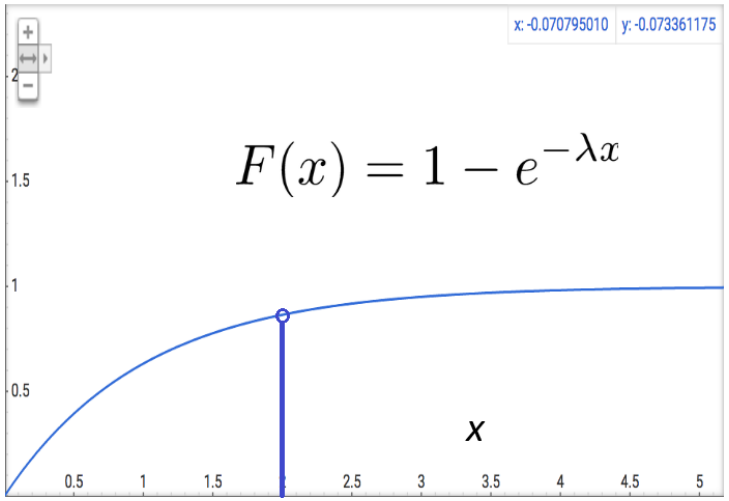



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Cumulative density function



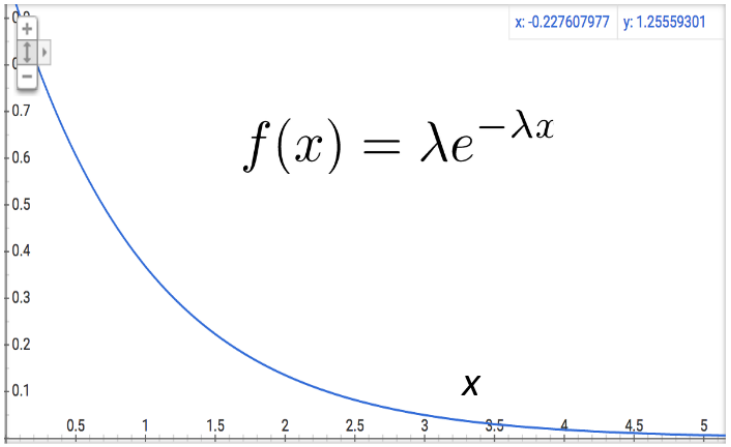

$$= F(2)$$

$$= 1 - e^{-2}$$

$$\approx 0.84$$

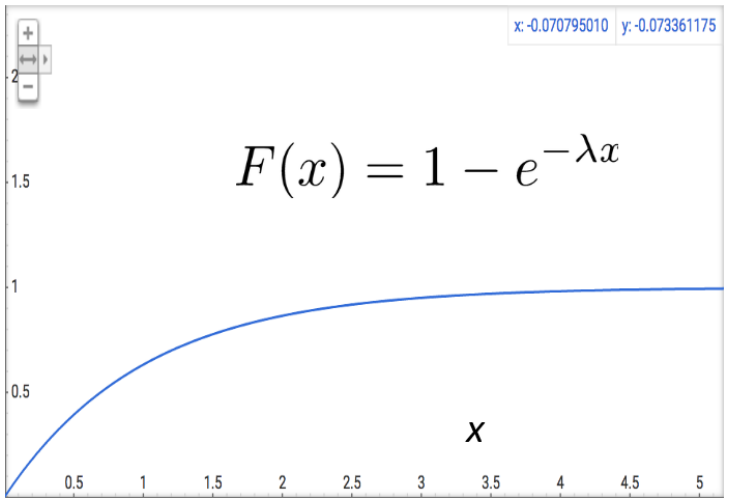
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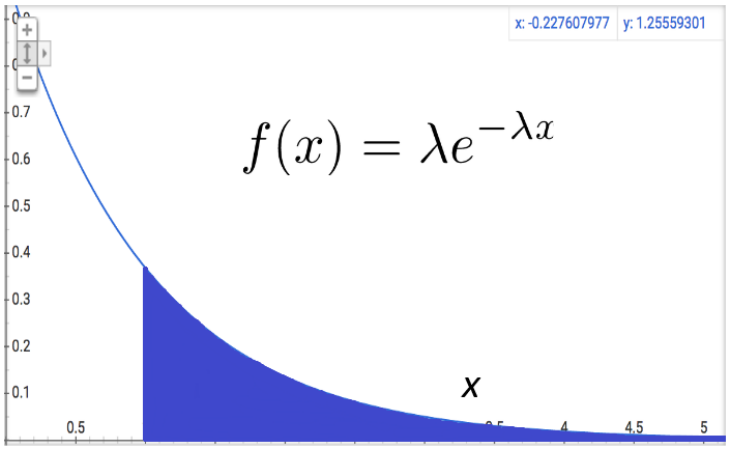

$P(X > 1)$

Cumulative density function



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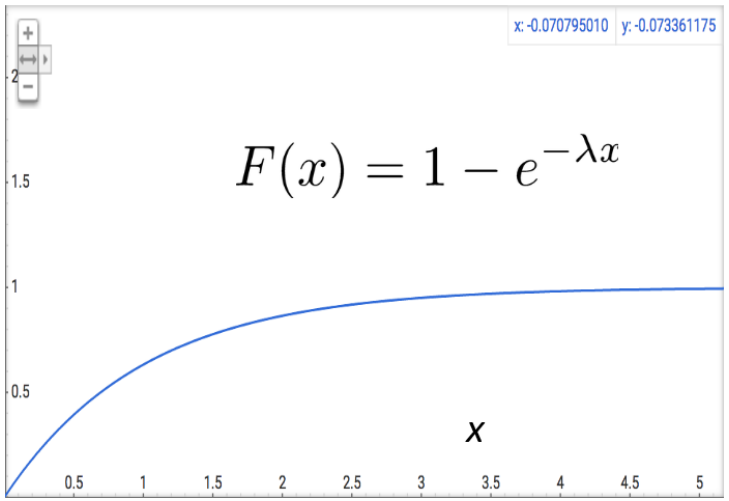
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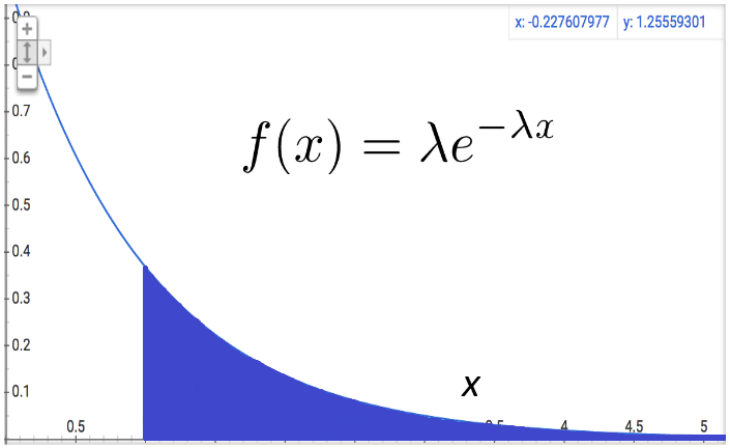
$$= \int_{z=1}^{\infty} f(x)\, dx$$

Cumulative density function



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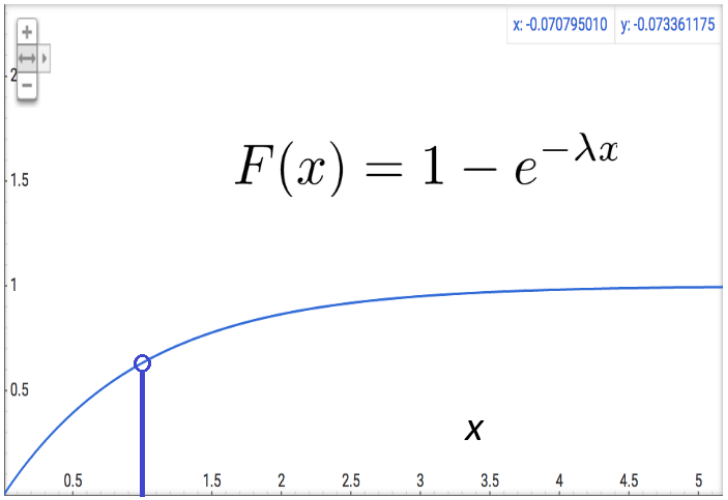



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OR

Cumulative density function



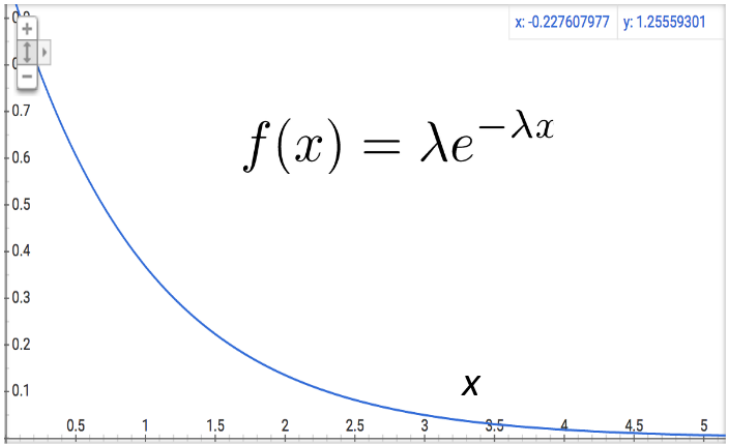

$$= 1 - F(1)$$

$$= e^{-1}$$

$$\approx 0.37$$

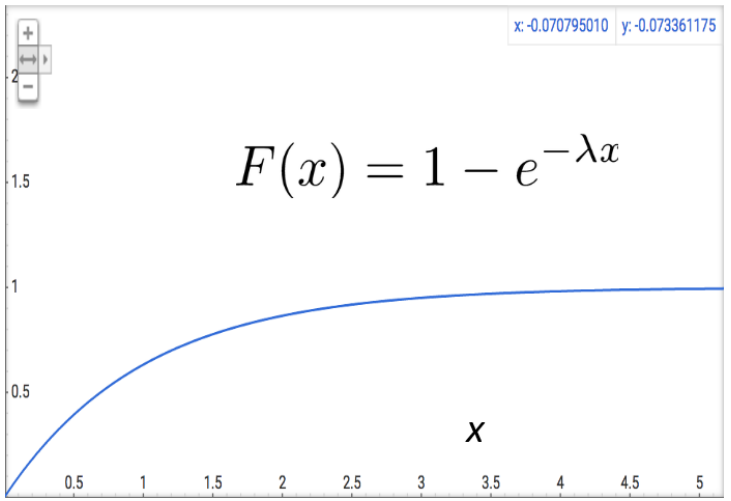
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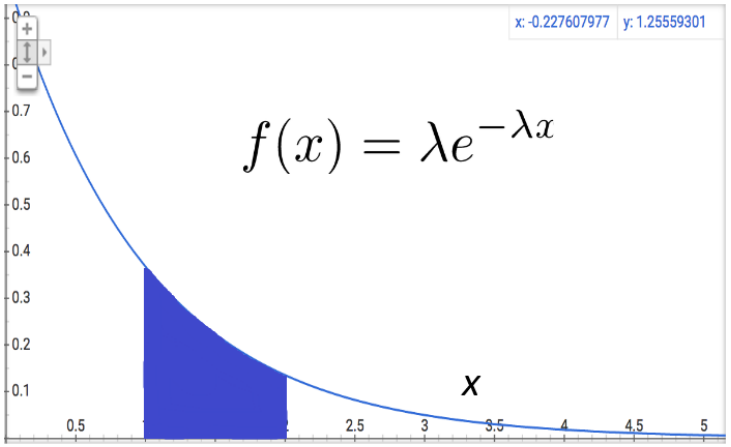

$$P(1 < X < 2)$$

Cumulative density function



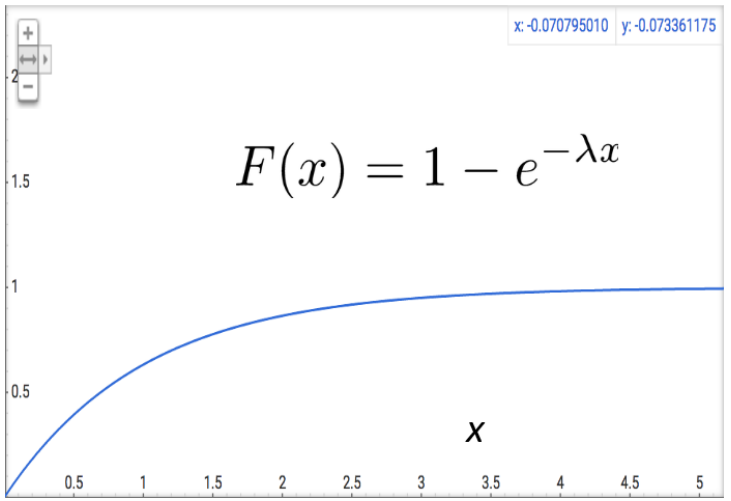
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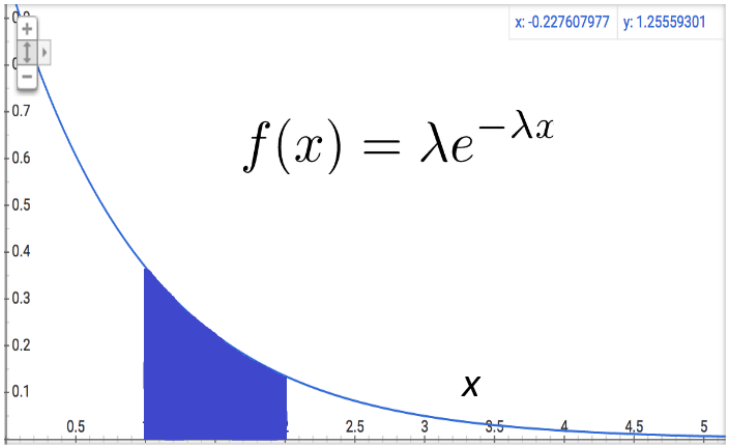

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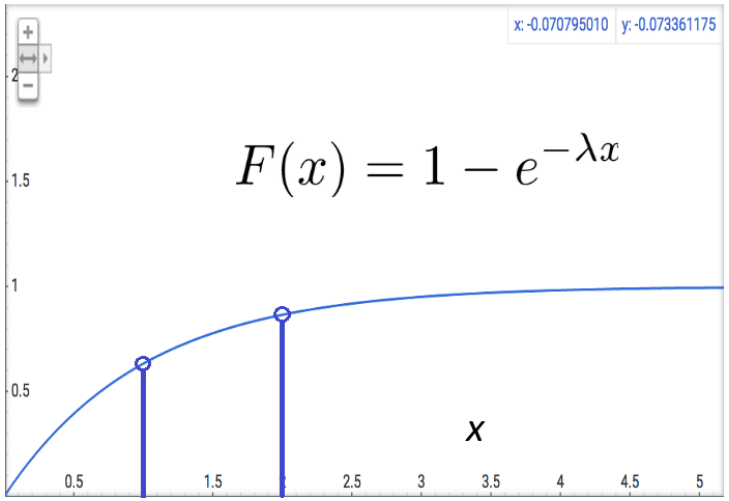
Probability density function




$$P(1 < X < 2) = \int_{x=1}^{2} f(x)\, dx$$

OR

Cumulative density function




$$= F(2) - F(1)$$

$$= \left(1 - e^{-2}\right) - \left(1 - e^{-1}\right)$$

$$\approx 0.23$$

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$$P(Y < 30) = F(30) = 1 - e^{-0.002 \times 30}$$

# Fun Fact: Exponential is Memoryless!

$$X \sim \text{Exp}(\lambda)$$
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$$P(X > s+t \mid X > s) = \frac{(X > s+t \text{and} X > s)}{P(X > s)}$$

$$= \frac{P(X > s+t)}{P(X > s)} \quad \text{Because } X > s+t \text{ implies } X > s$$

$$= \frac{1 - F(s+t)}{1 - F(s)} \quad \text{Def. of CDF}$$

$$= e^{\lambda t} \quad \text{Simplify}$$

$$= 1 - F(t) \quad \text{CDF of Exp}$$

$$= P(X > t) \quad \text{Def. of CDF}$$